TTech Application Guide for MSSP

1 Introduction

The purpose of this document is to provide an application programming guide for Application Providers (AP), such as banks, web shops, call center services, or online government services, providing services that require users to authenticate themselves or sign contracts with a mobile signature. A Mobile Signature Service Provider (MSSP) infrastructure in the mobile operator domain and a specific client application in the Application Provider (AP) domain are needed for this.

The mobile operator runs an MSSP Validation Server – MSSP's interface acts as a gateway for APs to request mobile / digital signatures from the end users.

The interface masks the AP from the complex background security systems and the mobile network technology. APs do not need to worry about different SIM cards, billing systems, Certification Authority (CA) hierarchies and certificate directories.

From the AP's perspective, connecting to MSSP Validation Server requires a fixed Internet connection and a client application that runs in the AP's environment. The client application is a customized application designed and implemented by the AP. This application is typically integrated with the AP's other systems. The AP must ensure that the client application meets the ETSI MSSP Standard [ETSI TS 102.204 v1.1.4] and implements the interfaces correctly.

MSSP Validation Server in turn is connected to the mobile operator's billing and message transmission systems. MSSP Validation Server forwards authentication and signature requests from the client application to the subscribers' mobile phones and vice versa over the mobile network. It handles possible error situations within the mobile network, and takes care of retry management.

This guide explains how to design applications to use the mobile operator's MSSP Validation Server platform. Note that; this document does not include the validation of the signature returned from the MSSP server. The validation of the signature should be done by retrieving the certificate from the CA.

1.1 Why an AP Needs Access to an MSSP

The MSS standards by ETSI define the interface to the mobile signature service, where the AP can ask the user to sign electronically some content with the mobile phone by sending a signature request to the end user's mobile operator.

The content to be signed can be, for example, an authentication challenge for logging into an on-line bank account, an order confirmation related to a purchase from a web shop, or any other content related to a transaction requiring the user's strong authentication and/or a verifiable commitment to an order.

1.2 TTech MSSP Client SDK

The AP's client application communicates with MSSP Validation Server by making SOAP 1.2 over HTTP web service calls. TTech MSSP Client SDK provides a higher level API that hides the complexity of making SOAP 1.2 over HTTP web service calls.

1.3 Overview of Roles

There are three main roles in the MSS system:

1. Mobile Subscriber (end user).

- A person with a mobile signature-enabled mobile telephone.

2. Application Provider (AP).

- Requires a digital signature from the Mobile Subscriber.

3. The MSSP platform at the mobile operator domain.

- Receives signature request from APs.
- Sends signature requests to the mobile subscriber.
- Receives and formats the signatures from the mobile subscriber.
- Retrieves the Mobile Subscriber's certificate. This function is optional; if a PKCS#1 signature is requested, there will be no certificate.
- Optionally validates the signature.
- Records all information in its transaction database.

2 Services for Application Providers

This chapter introduces the standard services available from the MSSP platform.

2.1 Three Types of Signature Services

This service implements the ETSI MSSP Mobile Signature Method. This method is used for the Mobile Signature Service and the Mobile Authentication Service.

The supported standard signature formats are:

- PKCS#7
- XML signature (RFC3275. XML-Signature Syntax and Processing.) [RFC3275]
- trmobilesignature:MultipleSignature: The “trmobilesignature:MultipleSignature” format is specific to Turkcell, which is an XML structure that contains PKCS#1 signatures and the end-user’s Certificate.

2.1.1 Mobile Signature Service

This service enables Application Providers to request non-repudiation signatures and to receive responses from the user. This service is used for signing contracts or transactions.

2.1.2 Profile Query Service

This service enables Application Providers to request end user’s certificate serial number and certificate hash which can be used in constructing the data to be signed.

2.1.3 Status Service

Application Providers can request and receive status information and signature (if available) of a previous Signature Request using an MSSP transaction ID. The status service is used for asynchronous communications between the client application and the MSSP platform.

3 How to Get Access to the MSSP Platform

1. Contact your mobile operator. You must probably sign a service agreement at this stage. Be prepared to provide the following information to the mobile operator:

- Description of the intended service.
- Address and location of the client application.
- In the asynchronous server-server mode, the Notification web service port URL for callback. You must also open your firewall for the callback.
- If you will be using Secure Sockets Layer (SSL), the MSSP must trust the Certificate Authority (CA) that issued your server certificate. The mobile operator must get that CA certificate.

2. Agree upon service charging.
3. Request from the mobile operator for the following information:
    - Application Provider ID (APID), which identifies you and your service among other services using the MSS.
    - Password for connecting to the MSS.
    - Test subscriptions. In practice, test subscriptions include a SIM card that you can use to test your service once it's up and running.
    - The IP address and port number to the MSS production environment.
4. If you are planning to use SSL, request an SSL client certificate from your mobile operator.
5. Read the SOAP 1.2 over HTTP web service call requirements in this document to specify your client application.
6. Write your client application in line with the programming instructions.
7. Test your service carefully. It is the AP's responsibility to ensure that the client application is tested properly and that its functionality is as expected. This also applies to error situations.
8. Launch your service and have your customers having a mobile signature-enabled SIM card take advantage of mobile signatures.

4 Connecting to MSSP Validation Service and the MSSP Platform

The AP communicates with MSSP Validation Service by making SOAP 1.2 over HTTP web service calls. A typical connection is protected by Secure Sockets Layer (SSL) to ensure privacy. The connection authenticates the AP through the use of an ID string, password, and possibly an SSL client certificate. For more information and support for SSL, contact your mobile operator.

The information needed to connect to MSSP Validation Service is presented in Table 1.

Table 1, MSSP connection information

Note: Mandatory information is marked with M and optional parameters are marked with O.

		Description
AP_ID	M	The ID of the AP
AP_PWD	M	The password that the AP uses to authenticate itself
SSL Certificate	O	The AP may have to request an SSL client certificate from the operator.
MSSP_ID	M	All APs including the MSSP are identified by means of URI-based identifier. The ID string identifying MSSP Validation Service: www.turkcelltech.com
WSDL file	O	The WSDL file describes the services offered by MSSP Validation Service
MSS_Signature URL	M	The URL of the SOAP endpoint that provides the MSS_Signature call. This URL is also contained in the WSDL file.
MSS_StatusQuery URL	O	The URL of the SOAP endpoint that provides the MSS_StatusQuery call This URL is also contained in the WSDL file.
MSS_ProfileQuery URL	O	The URL of the SOAP endpoint that provides the MSS_ ProfileQuery call This URL is also contained in the WSDL file.
SignatureProfile(s)	O	Signature Profile name that can be used with the MSSP Validation Service instance: Please see table below for possible signature profile values.

Table 2 Signature Profiles

Applet Type	SignatureProfile → mssURI element in MSS_SignatureReq	DataToBeSigned element in MSS_SignatureReq
Applet v1	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile1	Base64 encoded SHA-1 digest of data to be signed. (28 characters)
Applet v2	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile2	Base64 encoded SHA-256 digest of data to be signed. (44 characters)
Applet v2	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile2#sha1	Base64 encoded SHA-1 digest of data to be signed. (28 characters)
Applet v2	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile2#sha256	Base64 encoded SHA-256 digest of data to be signed. (44 characters)
Applet v2	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile2#sha384	Base64 encoded SHA-384 digest of data to be signed. (64 characters)
Applet v2	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile2#sha512	Base64 encoded SHA-512 digest of data to be signed. (88 characters)
Applet v2	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile2#other	Base64 encoded DER encoded PKCS#1 DigestInfo to sign.
Applet v3	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile3	- In binary Single Signing (Single Signature Request): If DataToBeSigned.MimeType=" application/octet-stream", data to be signed is a Base64 encoded SHA-256 digest. The digest consists of 44 characters. - In binary Multiple Signing (Multiple Signature Request): If DataToBeSigned.MimeType=" multipart/digest" and DataToBeSigned.Encoding="UTF-8", data to be signed consists of Base64 encoded SHA-256 digests which concatenated with semicolon(;). Each digest consists of 44 characters.
Applet v3	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile3#sha1	- In binary Single Signing (Single Signature Request): If DataToBeSigned.MimeType=" application/octet-stream",data to be signed is a Base64 encoded SHA-1 digest. The digest consists of 28 characters. - In binary Multiple Signing (Multiple Signature Request): If DataToBeSigned.MimeType=" multipart/digest" and DataToBeSigned.Encoding="UTF-8", data to be signed consists of Base64 encoded SHA-1 digests which concatenated with semicolon(;).Each digest consists of 28 characters.

Applet v3	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile3#sha256	- In binary Single Signing (Single Signature Request): If DataToBeSigned.MimeType=" application/octet-stream", data to be signed is a Base64 encoded SHA-256 digest. The digest consists of 44 characters. - In binary Multiple Signing (Multiple Signature Request): If DataToBeSigned.MimeType="multipart/digest" and DataToBeSigned.Encoding= "UTF-8", data to be signed consists of Base64 encoded SHA-256 digests which concatenated with semicolon(;). Each digest consists of 44 characters.
Applet v3	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile3#sha384	- In binary Single Signing (Single Signature Request): If DataToBeSigned.MimeType=" application/octet-stream", data to be signed is a Base64 encoded SHA-384 digest. The digest consists of 64 characters. - In binary Multiple Signing (Multiple Signature Request): If DataToBeSigned.MimeType="multipart/digest" and DataToBeSigned.Encoding= "UTF-8", data to be signed consists of Base64 encoded SHA-384 digests which concatenated with semicolon(;). Each digest consists of 64 characters.
Applet v3	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile3#sha512	- In binary Single Signing (Single Signature Request): If DataToBeSigned.MimeType=" application/octet-stream", data to be signed is a Base64 encoded SHA-512

		digest. The digest consists of 88 characters. - In binary Multiple Signing (Multiple Signature Request): If DataToBeSigned.MimeType=”multipart/digest” and DataToBeSigned.Encoding=”UTF-8”, data to be signed consists of Base64 encoded SHA-512 digests which concatenated with semicolon(;).Each digest consists of 88 characters.
Applet v3	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile3#other	- In binary Single Signing (Single Signature Request): If DataToBeSigned.MimeType=” application/octet-stream”, data to be signed is Base64 encoded DER encoded PKCS#1 `DigestInfo`. - In binary Multiple Signing (Multiple Signature Request): If DataToBeSigned.MimeType=”multipart/digest” and DataToBeSigned.Encoding=”UTF-8”, data to be signed consists of Base64 encoded DER encoded PKCS#1 ‘`DigestInfo’s` which concatenated with semicolon(;).

4.1 Integration

MSSP has a standard web service interface (SOAP), which means that it works with almost every language that supports SOAP, and every IDE supporting that language. Please, note that MSSP uses complex data types and RPC literals.

The WSDL file and associated schema files can be used to generate “Stub” source code. This code can then be called in order to contact MSSP Validation Service. You can also point your IDE to MSSP Validation Service address and generate “Stub” source code in that way.

For more on the WSDL and SOAP, refer to [MSSP].

4.2 SSL

If the connection to MSSP Validation Service uses TLS and SSL class 2 or 3, the AP must "trust" the instance of MSSP Validation Service it is connecting to. If the messaging mode is asynchronous server-server, MSSP Validation Service must also trust the Application Provider's server certificate.

5 Retrieving a Signature from MSSP Validation Service

This API of MSSP Validation Service can be used in three modes:

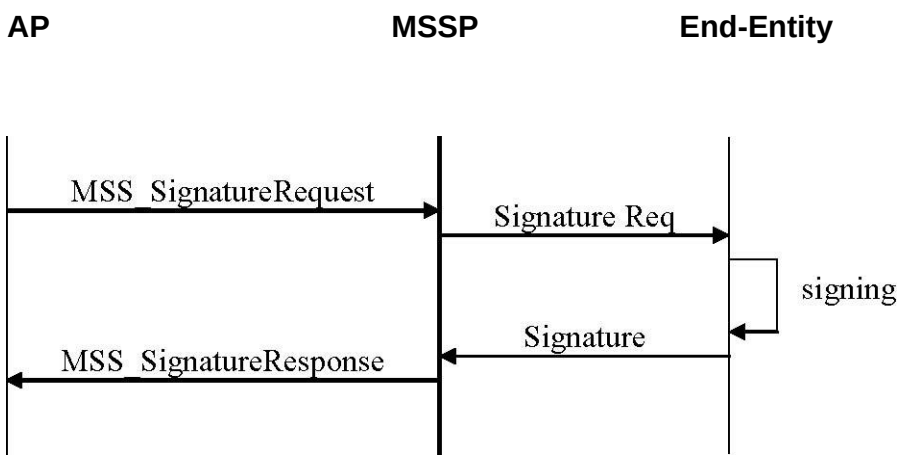
1. Synchronous mode (for testing only, not recommended for production)
2. Asynchronous client-server mode
3. Asynchronous server-server mode

The asynchronous client-server mode is also known as the polled mode and described in [MSSP]. The process of retrieving a signature from MSSP Validation Service is different depending on the server mode.

5.1 Synchronous Mode

In the synchronous mode, the AP initiates the signature request by calling MSS_Signature (see chapter 6.1), which then blocks until the signature has been received or the signing times out. The call returns the result of the signing, and validation if it was requested by the AP; and the signer's certificate if XML Signature is requested.

Figure 3, Getting a Signature Synchronously



5.2 Asynchronous Client-Server (Polled) Mode

In the polled mode, the AP initiates the signature request by calling MSS_Signature (see chapter 6.1), which immediately returns. The result can then be polled by using MSS_StatusQuery (see chapter 6.2).

When polling, we recommend you to make a break of approximately 30 seconds in the beginning between MSS_Signature and MSS_StatusQuery before the first MSS_StatusQuery request, because it usually takes at least 30 seconds for the user to receive and sign the request. After the break, the AP can poll responses every 10 seconds to receive the results using asynchronous calls.

5.3 Asynchronous Server-Server Mode

In the asynchronous server-server mode, the AP initiates the signature request by calling MSS_Signature (see chapter 6.1), which immediately returns. The AP must:

- Implement the actual SOAP service.
- Open the necessary firewall ports to allow MSSP access within the AP network.
- If you are planning to use SSL, remember that the MSSP must trust your server certificate. In practice, the certificate CN (Common Name) and your host name must match.

MSSP Validation Service sends an MSS_NotificationRequest (see chapter 6.5) to the AP when the signing process has been finished. The AP must respond with MSS_NotificationResponse (see chapter 6.6).

6 MSS Call Description

This section describes the calls that can be made to MSSP Validation Service. More detailed information about the parameters can be found in [MSSP], and other documents referenced by it.

6.1 MSS_Signature

The MSS_Signature operation takes an input of type MSS_SignatureRequest and responds with MSS_SignatureResponse.

6.1.1 MSS_SignatureRequest

The fields that make up MSS_SignatureRequest are shown in Table 3.

Table 3, MSS_SignatureRequest fields

Name	Type	Description	Req
MajorVersion	Integer	This value is always ‘1’.	Y
MinorVersion	Integer	This value is always ‘1’.	Y
AP_Info			Y
AP_ID	anyURI	ID of the AP. This value is assigned by the MSSP Validation Service operator.	Y
AP_PWD	String	Password of the AP. This value is assigned by the MSSP Validation Service operator.	Y
AP_TransID	NCName	A transaction number, created by the AP. This number uniquely identifies each transaction by this AP. The practical restrictions of NCName are that it cannot contain several symbol characters like :, @, $, %, &, /, +, ,, ;, whitespace characters or different parenthesis. Furthermore an NCName cannot begin with a number, dot or minus character although they can appear later in an NCName. We recommend using “_” (underscore) as the first character of NCName field. Length of this parameter can be maximum 32 characters. For detailed information about this field please refer to: http://www.w3.org/TR/1999/REC-xml-names-19990114/#NT-NCName	Y
Instant	DateTime	The date and time when the message was created. This value is used when a signature (hash) is constructed in addition to plain text. This prevents replay attacks.	Y
AP_URL	URL	A URL to the address that receives MSS_NotificationRequest from MSSP Validation Service when an AsynchronousServerServer operation has been finished.	N

Name	Type	Description	Req
MobileUser	MobileUserType		Y
MSISDN	String	The end-user's telephone number, for example 053234567890.	Y
DataToBeSigned	DataType	The data to be signed. The data can be - plain text, for Text Signing - a base64 encoded hash if the AP has done the hashing and is asking for the Binary Single Signing(Single Signature Request) - base64 encoded hashes which are concatenated with semicolon(;) for Binary Multiple Signing(Multiple Signature Request) In Single Signing(Single Signature Request), data to be signed is a single data (plain text or base64 encoded hash) and data length must be : - Max Lenght=223 for Text Signing (plain text) - Length = 28 for SHA1 binary signing (base64 encoded string length of 20 bytes) - Length = 44 for SHA256 binary signing (base64 encoded string length of 32 bytes) - Length = 64 for SHA384 binary signing (base64 encoded string length of 48 bytes) - Length = 88 for SHA512 binary signing (base64 encoded string length of 64 bytes) In Multiple Signing(Multiple Signature Request), data to be signed consists of multiple base64 encoded hashes. For each hash, length must be: - Length = 28 for SHA1 binary signing (base64 encoded string length of 20 bytes) - Length = 44 for SHA256 binary signing (base64 encoded string length of 32 bytes) - Length = 64 for SHA384 binary signing (base64 encoded string length of 48 bytes) - Length = 88 for SHA512 binary signing (base64 encoded string length of 64 bytes)	Y
value	String	The data-to-be-signed (DTBS)	Y
mime-type	String	The value of this attribute can be : - text/plain, for Text Signing - application/octet-stream, for Binary Single Signing(Single Signature Request) - multipart/digest, for multiple binary data(Multiple Signature Request)	Y
encoding	String	The value of this attribute can be UTF-8 or Base64. - UTF-8 for Text Signing - Base64 for Single Signing(Single Signature Request) - UTF-8 for Multiple Signing(Multiple Signature Request)	

Name	Type	Description	Req
DataToBeDisplayed	DataType	The text passed in this parameter is shown to the user before signing. - In Single Signing(Single Signature Request), the data is assumed to be a text. Max Length=223 - In Multiple Signing(Multiple Signature Request), the data is assumed to be multiple texts that are concatenated with semicolon(;). Each text corresponds to the digest in the same order in DataToBeSigned. Max Length =84 for each text.	N
value	String	The data-to-be-displayed (DTBD)	N
mime-type	String	The value of this attribute is text/plain	N
encoding	String	The value of this attribute is UTF-8	N
SignatureProfile	MssURIType	The signature profile to be used in the transaction processing. Details explained in Table 2.	Y
AdditionalServices	List of AdditionalService Type	There is one additional service that can be requested from MSSP. This field must contain following URIs for each of those services: http://uri.etsi.org/TS102204/v1.1.2#validate	N
MSS_Format	MssURI	This field indicates the format of the signature that should be created. The supported types are: - http://uri.etsi.org/TS102204/v1.1.2#PKCS7 is used in Single Signing(Single Signature Request) - http://uri.etsi.org/TS102204/v1.1.2#XML-Signature is used in Single Signing(Single Signature Request) - trmobilesignature:MultipleSignature, is used in Multiple Signing(Multiple Signature Request)	N

Name	Type	Description	Req
TimeOut	PositiveInteger	This field represents the time in seconds MSSP Validation Service waits before responding to the AP. In Single Signing(Single Signature Request), if TimeOut - is not set, timeout is set 300 seconds by application. defaultTimeout=300 - is less than minTimeout=30, timeout is set 30 seconds by application. - is greater than maxTimeout=600, timeout is set 600 seconds by application. In Multiple Signing(Multiple Signature Request), if TimeOut - is not set, it is calculated as below: TimeOut= defaultTimeout *documentCount defaultTimeout=60 e.g. if documentCount=3 TimeOut =60*3=180 sec - is less than minCombinedTimeout, it is calculated as below: TimeOut = minCombinedTimeOut= minTimeout *documentCount minTimeout=30 e.g. if documentCount=3 TimeOut =30*3=90 - is greater than maxCombinedTimeout, it is calculated as below: TimeOut = maxCombinedTimeout = maxTimeout *documentCount maxTimeout =120 e.g. if documentCount=3 TimeOut =120*3=360	N
ValidityDate	DateTime	Expiration date for the transaction. If this argument is not specified, a default timeout is used.	N
MessagingMode	MessagingModeType	Denotes the messaging mode. The value is synch, asynchClientServer, or asynchServerServer.	Y

An example Single Signature Request is shown below

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope"
xmlns:soap1="http://turkcelltech.com/mobilesignature/validation/soap" xmlns:v1="http://uri.etsi.org/TS102204/v1.1.2#">
   <soap:Header/>
   <soap:Body>
      <soap1:MSS_Signature>
         <MSS_SignatureReq MajorVersion="1" MessagingMode="synch" MinorVersion="1" TimeOut="120">
            <v1:AP_Info AP_ID="http://xxx.com.tr" AP_PWD="xxxx" AP_TransID="_12647510xxx" AP_URL=""
Instant="2009-12-22T08:23:38.778Z" xmlns:v1="http://uri.etsi.org/TS102204/v1.1.2#"/>
            <v1:MobileUser>
               <v1:MSISDN>053xxxxxxxx</v1:MSISDN>
            </v1:MobileUser>
            <v1:DataToBeSigned Encoding="Base64" MimeType="application/octet-stream"
xmlns:v1="http://uri.etsi.org/TS102204/v1.1.2#">+vLmZ1MH57M2GvI/+xDmrq7RKzg1Dv53YytK/822R+Q=</v1:DataToBeSigned>
            <v1:DataToBeDisplayed Encoding="UTF-8" MimeType="text/plain"
xmlns:v1="http://uri.etsi.org/TS102204/v1.1.2#">Test Single Signature</v1:DataToBeDisplayed>
            <v1:SignatureProfile>
               <v1:mssURI>http://uri.turkcellteknoloji.com.tr/MobileSignature/profile2#sha256</v1:mssURI>
            </v1:SignatureProfile>
            <v1:MSS_Format>
               <v1:mssURI></v1:mssURI>
            </v1:MSS_Format>
         </MSS_SignatureReq>
      </soap1:MSS_Signature>
   </soap:Body>
</soap:Envelope>
```

An example Multiple Signature Request is shown below

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope"
xmlns:soap1="http://turkcelltech.com/mobilesignature/validation/soap" xmlns:v1="http://uri.etsi.org/TS102204/v1.1.2#">
   <soap:Header/>
   <soap:Body>
      <soap1:MSS_Signature>
         <MSS_SignatureReq MajorVersion="1" MessagingMode="synch" MinorVersion="1" TimeOut="120">
            <v1:AP_Info AP_ID="http://xxx.com.tr" AP_PWD="xxxx" AP_TransID="_12647510xxx" AP_URL=""
Instant="2009-12-22T08:23:38.778Z" xmlns:v1="http://uri.etsi.org/TS102204/v1.1.2#"/>
            <v1:MobileUser>
               <v1:MSISDN>0533*******</v1:MSISDN>
            </v1:MobileUser>
            <v1:DataToBeSigned Encoding="UTF-8" MimeType="multipart/digest"
xmlns:v1="http://uri.etsi.org/TS102204/v1.1.2#">+vLmZ1MH57M2GvI/+xDmrq7RKzg1Dv53YytK/822R+Q=;+vLmZ1MH57M2GvI/+xDmrq7RK
zg1Dv53YytK/822R+Q=;+vLmZ1MH57M2GvI/+xDmrq7RKzg1Dv53YytK/822R+Q=</v1:DataToBeSigned>
            <v1:DataToBeDisplayed Encoding="UTF-8" MimeType="text/plain"
xmlns:v1="http://uri.etsi.org/TS102204/v1.1.2#">Signature1;Signature2;Signature3</v1:DataToBeDisplayed>
            <v1:SignatureProfile>
               <v1:mssURI>http://uri.turkcellteknoloji.com.tr/MobileSignature/profile3#sha256</v1:mssURI>
            </v1:SignatureProfile>
            <v1:MSS_Format>
               <v1:mssURI>trmobilesignature:MultipleSignature</v1:mssURI>
            </v1:MSS_Format>
         </MSS_SignatureReq>
      </soap1:MSS_Signature>
   </soap:Body>
</soap:Envelope>
```

6.1.1.1 Data and Signature combinations

The following parameters are required to get the end-user to sign plain text and to receive a PKCS#7 signature:

	Value
DataToBeSigned	
value	The string value should be user-readable text in UTF-8 format.
mime-type	text/plain
encoding	UTF-8
MSS_Format	http://uri.etsi.org/TS102204/v1.1.2#PKCS7

The following parameters are required to get the end-user to sign plain text and to receive an XML structure that contains a PKCS#1 signature and the end-user's certificate.

	Value
DataToBeSigned	
value	The string value should be user-readable text in UTF-8 format.
mime-type	text/plain
encoding	UTF-8
MSS_Format	http://uri.etsi.org/TS102204/v1.1.2#XML-Signature

The following table shows the parameters for binary Single Signing (Single Signature Request). These parameters are required to get the end-user to sign binary data (base64 encoded digest of the document) and to receive an XML structure that contains a PKCS#1 signature and the end-user's certificate. When binary data is signed the DataToBeDisplayed field is mandatory.

	Value
DataToBeSigned	
value	A base-64 encoded hash of the data to be signed. MSSP Validation Service (or the SIM cards) add the appropriate PKCS#1 structures before the data is encrypted by the end-user's private key
mime-type	application/octet-stream
encoding	Base64
DataToBeDisplayed	
value	The text shown to the user before signing.
mime-type	text/plain
encoding	UTF-8
MSS_Format	http://uri.etsi.org/TS102204/v1.1.2#XML-Signature

The following table shows the parameters for binary Multiple Signing (Multiple Signature Request). These parameters are required to get the end-user to sign binary data set and to receive an XML structure that contains PKCS#1 signatures and the end-user's certificate. The DataToBeDisplayed and MSS_Format fields are mandatory; DataToBeDisplayed must be a text set and MSS_Format must be "trmobilesignature:MultipleSignature".

	Value
DataToBeSigned	
value	Multiple Base-64 encoded hashes concatenated with semicolon(;). Each Base-64 hash represents the digest of a single data to be signed. MSSP Validation Service (or the SIM cards) add the appropriate PKCS#1 structures before the data is encrypted by the end-user's private key
mime-type	multipart/digest
encoding	UTF-8
DataToBeDisplayed	
value	The text shown to the user before signing. It consists of multiple texts that are concatenated with semicolon(;).
mime-type	text/plain
encoding	UTF-8
MSS_Format	trmobilesignature:MultipleSignature

The "trmobilesignature:MultipleSignature" format is specific to Turkcell, which is an XML structure that contains PKCS#1 signatures and the end-user's Certificate. An example is shown below.

```
<SignatureValue Id="Signature-05300415230-SIG">
MTYzNjJmMmFkZmNhMGY0YmU5NWZiMDA1ODJiMDdlYmQzNzVjMzRiN2U1M2U2YjRlMTUxZmIyMTdiZjlmMmNhNTcyMjhjNWZlMGI2Mjc5OWUxN2U1NDQ1NT
ViMGM5MjQzY2JmZmM5OWExMGVkNDgxZTYyNzRkNWY4N2I5ZjQyYTNiMmY1OWU0OWFkNDg0NzA2OTc5MTZmN2VmOTYzMmU0N2U1N2I0NWFkMGZjNjU5YzMx
YTc1YmUyZjViMWI3MGZhY2I0YjczZDhiNzhiYmI1Y2E0NDQxYjg0ZjQ3MGFhOTM3MTQwMTJjNGU2OTNhOGU2NmFlNzAyNDM5YmVmZGY5OTMwZjYwYjM3NG
Q4NTM0OGU3ZmNlZTEwNDRlNDAwZjk4MDE5MTc1MDg5YWIwOTVlMDVhYjViYzA5ZWFiYWRjYjMzYjM4OTQ0Njk3YjRlMzhiNmY3NDZjM2ZmM2M5ZTRjODA2
MGZhMjFmOGE4NjgzMDE0ODg2ZGQxNjI1OTU0OGY2MGZhYzBhMWJkOGMwZDY2NDAwZjE1YTliZDRjYmIxMmUwZGQzNDkxOWI0NTkzZWE4NmMwODAwOWFiYW
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</X509Certificate>
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The SignatureValue tag in trmobilesignature:MultipleSignature format includes a Base64 encoded structure that contains

- hex strings of binary signatures for successfully signed documents
- {responseCode:responseMessage} combinations(e.g. {208:EXPIRED_TRANSACTION}) for unsigned documents

The hex strings of the binary signatures and {responseCode:responseMessage} combinations are concatenated with semicolon(;). An example Base64 decoded value of SignatureValue tag is shown below. It includes three hex signatures and two {responseCode:responseMessage} combinations that are concatenated with semicolon(;)


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6.1.1.2 Additional Services

There is one service that can be requested from MSSP via AdditionalServices field of MSS_SignatureRequest web service. This field must contain following URIs for each of those requested services:

http://uri.etsi.org/TS102204/v1.1.2#validate

Validate Service:

When http://uri.etsi.org/TS102204/v1.1.2#validate value is present, the signature is validated using XMLDSig Core validation.

Please note that signature validation can be done by MSSP in some of the signing scenarios. Those scenarios are the ones that mime-type value in DataToBeSigned field is text/plain or text/xml.

6.1.2 MSS_SignatureResponse

The fields that make up MSS_SignatureResponse are shown on Table 4.

Table 4, MSS_SignatureResponse fields

Name	Type	Description	Req
MajorVersion	Integer	This value is always '1'.	Y
MinorVersion	Integer	This value is always '1'.	Y
AP_Info			Y
AP_ID	anyURI	ID of the AP. This value is assigned by the MSSP Validation Service operator.	Y
AP_PWD	String	Password of the AP. This value is assigned by the MSSP Validation Service operator.	Y
AP_TransID	NCName	A transaction number, created by the AP. This number must be unique for each transaction.	Y
Instant	DateTime	The date and time that were received from the AP.	Y
AP_URL	URL	A URL to the address that receives MSS_NotificationRequestfrom MSSP Validation Service when an AsynchronousServerServer operation has been finished.	N
MSSP_Info	MeshMemberType		Y
Instant	DateTime	The date and time when the message was created.	
DNSName	String	The DNS name of MSSP Validation Service if it is known by the AP. If it is not known, an empty value is also accepted.	Y
MobileUser	MobileUserType		Y
MSISDN	String	The end-user's telephone number, for example +3581234567890.	Y
UserIdentifier	String	If the feature is configured in MSSP Validation Service, this field can contain information read from the user certificate and mapped as it is configured in MSSP Validation Service.	N
Status	StatusType,	The current status of the transaction. For more information, see status codes and status messages in Table 5 below	Y
MSS_Signature	SignatureType,	The signature.	N
MSSP_TransID	NCName	A transaction number created by MSSP Validation Service for this transaction.	Y

Table 5, MSS_SignatureResponse Status

StatusCode	StatusMessage	Description
101	WRONG_PARAM	If a MSS_SignatureRequest parameter is assigned with a wrong value, WRONG_PARAM status is sent in MSS_SignatureResponse. Status message includes possible values. E.g. "WRONG_PARAM MessagingMode should be synch, asynchClientServer, or asynchServerServer."
102	MISSING_PARAM	If a required MSS_SignatureRequest parameter is empty, MISSING_PARAM status is sent in MSS_SignatureResponse. Status message includes missing parameter name. E.g. "MISSING_PARAM MobileUser.MSISDN is required."
103	WRONG_DATA_LENGTH	If a MSS_SignatureRequest parameter's length is not as expected, WRONG_DATA_LENGTH status is sent in MSS_SignatureResponse. Status message includes expected length. E.g. "WRONG_DATA_LENGTH DataToBeDisplayed.length longer than 223."
104	UNAUTHORIZED_ACCESS	If AP is not authorized UNAUTHORIZED_ACCESS status is sent in MSS_SignatureResponse.
105	UNKNOWN_CLIENT	If user is not a mobile signature subscriber, UNKNOWN_CLIENT status is sent in MSS_SignatureResponse. Status message includes client's MSISDN. E.g. "UNKNOWN_CLIENT 0533xxxxxxx"
107	INAPPROPRIATE_DATA	If a MSS_SignatureRequest parameter is assigned with an inappropriate value, INAPPROPRIATE_DATA status is sent in MSS_SignatureResponse. Status message includes appropriate values. E.g. "INAPPROPRIATE_DATA DataToBeSigned.MimeType: text/plain should be application/octet-stream."
108	INCOMPATIBLE_INTERFACE	If MSS_SignatureRequest.minorVersion or MSS_SignatureRequest.majorVersion is assigned with an inappropriate value, INCOMPATIBLE_INTERFACE status is sent in MSS_SignatureResponse. Status message includes appropriate values. E.g. "INCOMPATIBLE_INTERFACE MessageAbstractType.MinorVersion should be 1."
109	UNSUPPORTED PROFILE	If MSS_SignatureRequest.signatureProfile is assigned with an unsupported profile, UNSUPPORTED PROFILE status is sent in MSS_SignatureResponse. Status message includes appropriate values. E.g. "UNSUPPORTED PROFILE signatureProfile should be http://uri.turkcellteknoloji.com.tr/MobileSignature/profile2, http://uri.turkcellteknoloji.com.tr/MobileSignat

		ure/profile2#sha256, http://uri.turkcellteknoloji.com.tr/MobileSignature/profile2#sha384, http://uri.turkcellteknoloji.com.tr/MobileSignature/profile2#sha512 or http://uri.turkcellteknoloji.com.tr/MobileSignature/profile2#other”
208	EXPIRED_TRANSACTION	- In Single Signing: If timeout occurs, signing process is expired. - In Multiple Signing: If timeout occurs, it doesn’t mean that all the documents are unsigned because of timeout. Some documents might be singed before timeout. Response for each document should be decoded from MSS_Signature field of MSS_SignatureResponse . - See the TimeOut details on MSS_SignatureRequest.TimeOut row of Table 3 (MSS_SignatureRequest fields)
401	USER_CANCEL	Signing is canceled by user.
404	NO_KEY_FOUND	Key not found. Customer should contact with Turkcell.
422	NO_CERT_FOUND	User certificate not found.
425	ERROR_CERTIFICATE	User Certificate is invalid. User Certificate may be expired.
500	SIGNATURE	Single Signing is successful. The signature can be obtained from MSS_SignatureResponse .MSS_Signature field. See the “6.1.1.1 Data and Signature combinations” section to check over the MSS_Format for Single Signing that forms the MSS_Signature.Base64Signature field.
501	MULTIPLE_SIGN_COMPLETED	Multiple Signing is completed. Please note that MULTIPLE_SIGN_COMPLETED doesn’t mean all the documents are signed successfully. It only indicates that the multiple signing process is completed successfully. So, each document has a signing result and the signing results for all documents can be obtained from MSS_SignatureResponse .MSS_Signature field. See the “6.1.1.1 Data and Signature combinations” section to check over the trmobilesignature:MultipleSignature value for MSS_Format field that forms the MSS_SignatureResponse.MSS_Signature.Base64Signature field.
502	VALID_SIGNATURE	If verification is enabled, signature is verified and signature is valid.
503	INVALID_SIGNATURE	If verification is enabled, signature is verified and signature is invalid.
504	OUTSTANDING_TRANSACTION	Signing process is uncompleted.
900	SYSTEM_ERROR	A system level error occurred. E.g. SYSTEM_ERROR weblogic.jms.common.LostServerException [JMSClientExceptions:055169]Network connection to server was lost.

An example Single Signature Response is shown below. Base64Signature tag includes a Base64 encoded string for XML response structure. Please see the “6.1.1.1 Data and Signature Combinations” section to check over the http://uri.etsi.org/TS102204/v1.1.2#XML-Signature MSS_Format for binary Single Signing (Single Signature Request).

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope">
   <soap:Body>
      <ns1:MSS_SignatureResponse xmlns:ns1="http://turkcelltech.com/mobilesignature/validation/soap">
         <MSS_SignatureResp MSSP_TransID="_655324" MajorVersion="1" MinorVersion="1"
xmlns:ns2="http://www.w3.org/2000/09/xmldsig#" xmlns:ns3="http://www.w3.org/2003/05/soap-envelope"
xmlns:ns4="http://www.w3.org/2001/04/xmlenc#" xmlns:ns5="http://uri.etsi.org/TS102204/v1.1.2#">
            <ns5:AP_Info AP_ID="http://xxx.com.tr" AP_TransID="_12647*****" AP_PWD="xxx123"
Instant="2009-12-22T08:23:38.778Z" AP_URL=""/>
            <ns5:MSSP_Info Instant="2020-02-16T18:00:11.208+02:00">
               <ns5:MSSP_ID/>
            </ns5:MSSP_Info>
            <ns5:MobileUser>
               <ns5:MSISDN>0533******</ns5:MSISDN>
            </ns5:MobileUser>
            <ns5:MSS_Signature>
<ns5:Base64Signature>PFNpZ25hdHVyZVZhbHVlIElkPSJTaWduYXR1cmUtMDUzMzE1MjQ2ODctU0lHIj5CZDNmc2VFOURDSEtBdzhzZXRrVHBSd3Zqe
VBUV042RENscHNMYkFQbjA0MCtkdXg1OU9KNmpTd29vcFpma1hlCmh1dmIwT0grN3ppSzFYZ1EwY0cwZU1aZXkvYjE2SkhDSlZFSnhwMWtNWkYzQm14cWl
oSWJBM2FuQmZUNmJYNXAKdHRNS2pCYmIzZmdoNTlWNkpDdzN0Yjl6dFBrMk1OejljJdXdPc1FkaFJBZnJ2SC9wOGJ6N2thSHNCYUJYU2I2MQpSRTZmWTVVT
1d4YmVmS0xNLzE3TWs4TVJkZDVJWUhiMG5CL2F4ZlNQMGVPTzdjcEFZQW5KU1VuM21QY1NSbU1lCjFRNlRqRmg0T3IzTjRxUnlsUWJOREIxVFhVVUJUSGx
zQTR4Q0Y5MXhwY3luenJURDhoT0hXYXk3NDdXNXNkM2UKOVJNT0ZzU2RGTmZGNEgrSTRHOUYzQT09PC9TaWduYXR1cmVWYWx1ZT48S2V5SW5mbz4KPEtle
VZhbHVlPgo8UlNBS2V5VmFsdWU+CjxNb2R1bHVzPkFKVFZDTklzUGxwQmpTZ0IyRTN0aVJXa1c4RDQrSzRzV0FXV0FMejAwL2Z0WG4yTXl3d2srODNUWHN
HWkZnaXIKczNPWWR1MXB4c1JVOW1VWjB5ODBLbVNHWGpxWElPdkNnUVF1OUxXRXBDVjhlRHpvQkhiZWd0Zys2ODVkR202bApOZ2RFNkg2SEJxTUxFQVkvR
XBxa29IOUx0UXRoblNqeHRLd21DOTZTRUZWT0wwUFhJMHRtazVTbzExTFo0L09nCkFEL01OVWtqdDRVVmNLaEw4Ym1uZmZhM3kyYmtCblF5TmJuRnhVbkx
HYlZtS3FHcUFOYkJ0dExKMWczYTR2dHcKMkd0RC9ZU1o2RUNuZW1GVWE4TjBJYk15WHZqR1lsSzFOQXdwNDZadDdtelQ1REszUzR3dTFvTDJyeEl1UmJ5T
go0ZW1EWDcrSHRuc09nUFlkV2lqcklwRT08L01vZHVsdXM+CjxFeHBvbmVudD5BUUFCPC9FeHBvbmVudD4KPC9SU0FLZXlWYWx1ZT4KPC9LZXlWYWx1ZT4
KPFg1MDlEYXRhPjxYNTA5Q2VydGlmaWNhdGU+TUlJR3BUQ0NCWTJnQXdJQkFnSVJBTHh4ZjkwZm5Bcmw2YzZNTitud1ZQc3dEUVlKS29aSWh2Y05BUUVMQ
lFBdwpiekVMTUFrR0ExVUVCaE1DVkZJeEtEQW1CZ05WQkFvTUgwVnNaV3QwY205dWFXc2dRbWxzWjJrZ1IzVjJaVzVzCmFXZHBJRUV1VXk0eE5qQTBCZ05
WQkFNTUxWUkZVMVFnVkhWeWEyTmxiR3dnVFc5aWFXd2dUa1ZUSUVocGVtMWwKZENCVFlXZHNZWGxwWTJsemFTQlRNakF1RnceU1EQXhNak14TWpNeU1ET
mFGdzB5TVRBeE1qSXhNak15TUROYQpNRW94Q3pBSkJnTlZCQVlUQWxSU01SRXdEd1lEVlFRS0RBaEdhWEpsSUV4TVZERVVNQklHQTFVRUJSTUxOelkxCk5
ETTBOVFkzTmpVeEVqQVFCZ05WQkFNTUNVMWxjblFnU1c1aGJqQ0NBU0l3RFFZSktvWklodmNOQVFFQkJRUQKZ2dFUEFEQ0NBUW9DZ2dFQkFKVFZDTklzU
GxwQmpTZ0IyRTN0aVJXa1c4RDQrSzRzV0FXV0FMejAwL2Z0WG4yTQp5d3drKzgzVFhzR1pGZ2lyczNPWWR1MXB4c1JVOW1VWjB5ODBLbVNHWGpxWElPdkN
nUVF1OUxXRXBDVjhlRHpvCkJIYmVndGcrNjg1ZEdtNmxOZ2RFNkg2SEJxTUxFQVkvRXBxa29IOUx0UXRoblNqeHRLd21DOTZTRUZWT0wwUFgKSTB0bWs1U
28xMUxaNC9PZ0FEL01OVWtqdDRVVmNLaEw4Ym1uZmZhM3kyYmtCblF5TmJuRnhVbkxHYlZtS3FHcQpBTmJCdHRMSjFnM2E0dnR3Mkd0RC9ZU1o2RUNuZW1
GVWE4TjBJYk15WHZqR1lsSzFOQXdwNDZadDdtelQ1REszClM0d3Uxb0wycnhJdVJieU40ZW1EWDcrSHRuc09nUFlkV2lqcklwRUNBd0VBQWFPQ0ExOHdnZ
05iTUVJR0NDc0cKQVFVRkJ3RUJCRFl3TkRBeUJnZ3JCZ0VGQlFjd0FZWW1hSFIwY0RvdkwzUmxjM1J2YWTNOd01pNWxMV2QxZG1WdQpMbU52YlM5dlkzTnd
MbmgxWkdFd0h3WURWUjBqQkJnd0ZvQVVUOWdTYXpBZlFyblpydUlxOStkSkZaN0U5T1V3CmdnRnlCZ05WSFNBRWdnRnBNSUlCWlRDQnNRWUdZSVlZQXdBQ
k1JR21NRFlHQ0NzR0FRVUZCd0lCRmlwb2RIUncKT2k4dmQzZDNMbVV0WjNWMlpXNHVZMjl0TDJSdlkzVnRaVzUwY3k5T1JWTlZSUzV3WkdZd2JBWUlLd1l
CQlFVSApBZ0l3WUJwZVFuVWdjMlZ5ZEdsbWFXdGhMQ0ExTURjd0lITmhlY1N4Yk1TeE1FVnNaV3QwY205dWFXc2d4TEJ0CmVtRWdTMkZ1ZFc1MWJtRWdaO
E8yY21VZ2JtbDBaV3hwYTJ4cElHVnNaV3QwY205dWFXc2djMlZ5ZEdsbWFXdGgKWk1TeGNqQ0JyZ1lKWUlZWUF3QUJBUUVETUlHZ01EY0dDQ3NHQVFVRkJ
3SUJGaXRvZEhSd09pOHZkM2QzTG1VdApaM1YyWlc0dVkyOXRMMlJ2WTNWdFpXNTBjeTlOUzA1RlUwa3VjR1JtTUdVR0NDc0dBUVVGQndJQ01Ga2FWMEoxC
klITmxjblJwWm1scllTd2dUVXRPUlZOSklHdGhjSE5oYmNTeGJtUmhJSGxoZWNTeGJteGhibTNFc2NXZklHSnAKY2lCdWFYUmxiR2xyYkdrZ1pXeGxhM1J
5YjI1cGF5QnpaWEowYVdacGEyRmt4TEZ5TGpCZEJnTlZIUjhFVmpCVQpNRktnVUtCT2hreG9kSFJ3T2k4dmRHVnpkSE5wYkM1bExXZDFkbVZ1TG1OdmJTO
UZiR1ZyZEhKdmJtbHJRbWxzCloybEhkWFpsYm14cFoybEJVMDFMVGtWVFNTMVRNaTlNWVhSbGMzUkRVa3d1WTNKc01BNEdBMVVkRHdFQi93UUUKQXdJR3d
EQ0Jnd1lJS3dZQkJRVUhBUU1FZHpCMU1BZ0dCZ1FBamtZQkFUQnBCZ3RnaGhnQlBRQUJwMDRCQVF4YQpRblVnYzJWeWRHbG1hV3RoTENBMU1EY3dJSE5oZ
Vdsc2FTQkZiR1ZyZEhKdmJtbHJJRWx0ZW1FZ1MyRnVkVzUxCmJtRWdaOE8yY21VZ2JtbDBaV3hwYTJ4cElHVnNaV3QwY205dWFXc2djMlZ5ZEdsbWFXdGh
aR2x5TUZBR0ExVWQKQ1FSSk1FY3dGQVlJS3dZQkJRVUhDUUl4Q0FRR1FXNXJZWEpoTUIwR0NDc0dBUVVGQndrQk1SRVlEekU1TnpneApNak14TWpBd01EQ
XdXakFRQmdnckJnRUZCUWNKQkRFRUJBSlVVakFZQmdOVkhSRUVFVEFQZ1ExbWFYSmxRR1pwCmNtNtVXVZMjl0TUIwR0ExVWREZ1FXQkJTTUEwWGRTUG1obHN
rZlQ1VThHZi93SmVoMm1UQU5CZ2txaGtpRzl3MEIKQVFzRkFBT0NBUUVBZDkxKzNZNjRlQ3lickNVZjNyN0drVEJ6NTJBdzdxNW9PU3lpcFVNRmwzZWpsV
2FabVJVTQo3QXpsYTVNanYrRmttTXJsWFgzWGRzR2JPSlUxSTI3bDRaYlBlc1VsL3RSbUplQzJSVXkzMEJ5ckl0UzdxV29NClluMG5oeE8zdzEySEJXOHZ
ZdTZtWlE3Ym5Jek8rczgyUHZOUFhVTWJwWGVTVDE1N3YwZ3pSQXEyd0x0ckhVYzMKRlRPWWhZeHM3SWZmZzVmb3Q2VkFDU0tpdEZFYUlqYjBPUE9ZWTZCa
TJxbUcyZkVaaDFiT21xWFgvejd1b3R3MwpmV1hXS3dvTnJmYWZrNXVSMVhVTksyd0p2UjB6QldUMXMyWCtmMzluMjdpVElIaXQyNjMrVkF4SXEyRUN6Zm9
JCkVEelV4WXhPM1dNb0l0OW9YNWsrdTFrTFVaWDRYL3lnclE9PTwvWDUwOUNlcnRpZmljYXRlPjwvWDUwOURhdGE+PC9LZXlJbmZvPg==</ns5:Base64S
ignature>
            </ns5:MSS_Signature>
            <ns5:SignatureProfile>
               <ns5:mssURI>http://uri.turkcellteknoloji.com.tr/MobileSignature/profile2</ns5:mssURI>
            </ns5:SignatureProfile>
            <ns5:Status>
               <ns5:StatusCode Value="500"/>
               <ns5:StatusMessage>SIGNATURE</ns5:StatusMessage>
            </ns5:Status>
         </MSS_SignatureResp>
      </ns1:MSS_SignatureResponse>
   </soap:Body>
</soap:Envelope>
```

An example Multiple Signature Response is shown below. Base64Signature tag includes a Base64 encoded string for XML response structure. Please see the “6.1.1.1 Data and Signature Combinations” section to check over the XML

response structure(MSS_Format= trmobilesignature:MultipleSignature) for binary Multiple Signing (Multiple Signature Request).


```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope">
   <soap:Body>
      <ns1:MSS_SignatureResponse xmlns:ns1="http://turkcelltech.com/mobilesignature/validation/soap">
         <MSS_SignatureResp MSSP_TransID="_65650" MajorVersion="1" MinorVersion="1"
xmlns:ns2="http://www.w3.org/2000/09/xmldsig#" xmlns:ns3="http://www.w3.org/2003/05/soap-envelope"
xmlns:ns4="http://www.w3.org/2001/04/xmlenc#" xmlns:ns5="http://uri.etsi.org/TS102204/v1.1.2#">
            <ns5:AP_Info AP_ID="http://xxx.com.tr" AP_TransID="_126*******" AP_PWD="xxx123"
Instant="2009-12-22T08:23:38.778Z" AP_URL=""/>
            <ns5:MSSP_Info Instant="2020-02-16T18:03:35.428+02:00"><ns5:MSSP_ID/></ns5:MSSP_Info>
            <ns5:MobileUser>
               <ns5:MSISDN>0533*******</ns5:MSISDN>
            </ns5:MobileUser>
            <ns5:MSS_Signature>
<ns5:Base64Signature>PFNpZ25hdHVyZVZhbHVlIElkPSJTaWduYXR1cmUtMDUzMzE1MjQ2ODctU0lHIj5NRFZrWkdSbVlqRmxNVE5rTUdNeU1XTmhNR
E13WmpKak4yRmtPVEV6WVRVeFl6Sm1PR1l5TTJRek5UaGtaVGd6Ck1HRTFZVFpqTW1SaU1EQm1PV1kwWlRNMFpqbGtZbUl4WlRka016ZzVaV0V6TkdJd1l
USTRZVFU1TjJVME5XUmwKT0RabFltUmlaREJsTVdabFpXWXpPRGhoWkRVM09ERXdaREZqTVdJME56aGpOalZsWTJKbU5tWTFaVGc1TVdNeQpNalUxTVRBN
Vl6WTVaRFkwTXpFNU1UYzNNRFkyWXpaaE9HRXhNakZpTURNM05tRTNNRFZtTkdaaE5tUTNaVFk1CllqWmtNekJoT0dNeE5tUmlaR1JtT0RJeFpUZGtOVGR
oTWpReVl6TTNZalZpWmpjellqUm1PVE0yTXpCa1kyWmsKTWpKbFl6QmxZakV3TnpZeE5EUXdOMlZpWW1NM1ptVTVaakZpWTJaaU9URmhNV1ZqTURWaE1EV
TNORGxpWldJMQpORFEwWlRsbU5qTTVOVEJsTldJeE5tUmxOMk5oTW1OalptWTFaV05qT1ROak16RXhOelZrWlRRNE5qQTNObVkwCk9XTXhabVJoWXpWbU5
EaG1aREZsTXpobFpXUmpZVFF3TmpBd09XTTVORGswT1dZM09UaG1OekV5TkRZMk16RmwKWkRVd1pUa3pPR00xT0RjNE0yRmlaR05rWlRKaE5EY3lPVFV3T
m1Oa01HTXhaRFV6TldRME5UQXhOR00zT1RaagpNRE00WXpReU1UZGtaRGN4WVRWalkyRTNZMlZpTkdNelpqSXhNemczTlRsaFkySmlaVE5pTldJNVlqRmt
aR1JsClpqVXhNekJsTVRaak5EbGtNVFJrTjJNMVpUQTNaamc0WlRBMlpqUTFaR003TURWa1pHUm1ZakZsTVROa01HTXkKTVdOaE1ETXdaakpqTjJGa09UR
XpZVFV4WXpKbU9HWXlNMlF6TlRoa1pUZ3pNR0UxWVRaak1tUmlNREJtT1dZMApaVE0wWmpsa1ltSXhaVGRrTXpnNVpXRXpOR0l3WVRJNFlUVTVOMlUwTld
SbE9EWmxZbVJpWkRCbE1XWmxaV1l6Ck9EaGhaRFUzT0RFd1pERmpNV0kwTnpoak5qVmxZMkptTm1ZMVpUZzVNV015TWpVMU1UQTVZelk1WkRZME16RTUKT
VRjM01EWTJZelpoT0dFeE1qRmlNRE0zTm1FM01EVm1OR1poTm1RM1pUWTVZalprTXpCaE9HTXhObVJpWkdSbQpPREl4WlRka05UZGhNalF5WXpNM1lqVml
aamN6WWpSbU9UTTJNekJrWTJaa01qSmxZekJsWWpFd056WXhORFF3Ck4yVmlZbU0zWm1VNVpqRmlZMlppT1RGaE1XVmpNRFZoTURVM05EbGlaV0kxTkRRM
FpUbG1Oak01TlRCbE5XSXgKTm1SbE4yTmhNbU5qWm1ZMVpXTmpPVE5qTXpFeE56VmtaVFE0TmpBM05tWTBPV014Wm1SaFl6Vm1ORGhtWkRGbApNemhsWld
SallUUXdOakF3T1dNNU5EazBPV1kzT1RobU56RXlORkFkyTXpGbFpEVXdaVGt6T0dNMU9EYzRNMkZpClpHTmtaVEpoTkRjeU9UVXdObU5rTUdNeFpEVXpOV
1EwTlRBeE5HTTNPVFpqTURNNFl6UXlNVGRrWkRjeFlUVmoKWTJFM1kyVmlOR016WmpJeE16ZzNOVGxoWTJKaVpUTmlOV0k1WWpGa1pHUmxaalV4TXpCbE1
UWmpORGxrTVRSawpOMk0xWlRBM1pqZzRaVEEyWmpRMVpHTT08L1NpZ25hdHVyZVZhbHVlPjxLZXlJbmZvPgo8S2V5VmFsdWU+CjxSU0FLZXlWYWx1ZT4KP
E1vZHVsdXM+QUpUVkNOSXNQbHBCalNnQjJFM3RpUldrVzhENCtLNHNXQVdXQUx6MDAvZnRYbjJNeXd3ays4M1RYc0daRmdpcgpzM09ZZHUxcHhzUlU5bVV
aMHk4MEttU0dYanFYSU92Q2dRUXU5TFdFcENWOGVEem9CSGJlZ3RnKzY4NWRHbTZsCk5nZEU2SDZIQnFNTEVBWS9FcHFrb0g5THRRdGhuU2p4dEt3bUM5N
lNFRlZPTDBQWEkwdG1rNVNvMTFMWjQvT2cKQUQvTU5Va2p0NFVWY0toTDhibW5mZmEzeTJia0JuUXlOYm5GeFVuTEdiVm1LcUdxQU5iQnR0TEoxZzNhNHZ
0dwoyR3REL1lTWjZFQ25lbUZVYThOMEliTXlYdmpHWWxLMU5Bd3A0Nlp0N216VDVESzNTNHd1MW9MMnJ4SXVSYnlOCjRlbURYNytIdG5zT2dQWWRXaWpyS
XBFPTwvTW9kdWx1cz4KPEV4cG9uZW50PkFRQUI8L0V4cG9uZW50Pgo8L1JTQUtleVZhbHVlPgo8WDUwOURhdGE+PFg1MDlDZXJ0aWZpY2F0ZT5NSUlHcFRDQ0ZJZmmdBd0lCQWdJUkFMeHhmOTBmbkFybDZjNk1OK253VlBzd0RRWUpLb1pJaHZjTkFRRUxCUUF3CmJ6RUxNQWtHQTFVRUJoTUNWR
k14S0RBbUJnTlZCQW9NSDBWc1pXdDBjbTl1YVdzZ1FtbHNaMmtnUjNWMlpXNXMKYVdkcElFRXVVeTR4TmpBMEJnTlZCQU1NTFZSRlUxUWdWSFZ5YTJObGJ
Hd2dUVzlpYVd3Z1RrVlRJRWhwZW0xbApkQ0JUWVdkc1lYbHBZMmx6YVNCVE1qQWVGdzB5TURBeE1qTXhNak15TUROYUZ3MHlNVEF4TWpJeE1qTXlNRE5hC
k1Fb3hDekFKQmdOVkJBWVRBbFJTTVJFd0R3WURWUVFLREFoR2FYSmxJRXhNVkRFVU1CSUdBMVVFQlJNTE56WTEKTkRNME5UWTNOalV4RWpBUUJnTlZCQU1
NQ1UxbGNuUWdTVzVoYmpDQ0FTSXdEUVlKS29aSWh2Y05BUUVCQlFBRApnZ0VQQURDQ0FRb0NnZ0VCQUpUVkNOSXNQbHBCalNnQjJFM3RpUldrVzhENCtLN
HNXQVdXQUx6MDAvZnRYbjJNCnl3d2srODNUWHNHWkZnaXJzM09ZZHUxcHhzUlU5bVVaMHk4MEttU0dYanFYSU92Q2dRUXU5TFdFcENWOGVEem8KQkhiZWd
0Zys2ODVkR202bE5nZEU2SDZIQnFNTEVBWS9FcHFrb0g5THRRdGhuU2p4dEt3bUM5NlNFRlZPTDBQWApJMHRtazVTbzExTFo0L09nQUQvTU5Va2p0NFVWY
0toTDhibW5mZmEzeTJia0JuUXlOYm5GeFVuTEdiVm1LcUdxCkFOYkJ0dExkMWczYTR2dHcyR3REL1lTWjZFQ25lbUZVYThOMEliTXlYdmpHWWxLMU5Bd3A
0Nlp0N216VDVESzMKUzR3dTFvTDJyeEl1UmJ5TjRlbURYNytIdG5zT2dQWWRXaWpySXBFQ0F3RUFBYU9DQTE4d2dnTmJNRUlHQ0NzRwpBUVVGQndFQkJEW
XdOREF5QmdnckJnRUZCUWN3QVlZbWFIUjBjRG92TDNSbGMzUnZZM053TWk1bExxXZDFkbVZ1CkxtTnZiUzl2TWNOd0xuaDFaR0V3SHdZRFZSMGpCQmd3Rm9
BVVQ5Z1NhekFmUXJuWnJ1SXE5K2RKRlo3RTlPVXcKZ2dGeUJnTlZIU0FFZ2dGcE1JSUJaVENCc1FZR1lJWVlBd0FCTUlHbU1EWUdDQ3NHQVFVRkJ3SUJGa
XBvZEhSdwpPaTh2ZDNkM0xtVXRaM1YyWlc0dVkyOXRMMlJ2WTNWdFpXNTBjeTlPUlZOVlJTNXdaR1l3YkFZSUt3WUJCUVVlCepYSjBhV1pwYTJGa3h
MRnlMakJkQmdOVkhSOEVWakJVCk1GS2dVS0JPaGt4b2RIUndPaTh2ZEdWemRITnBiQzVsTFdkMWRtVnVMbU52Y1M5RmJHVnJkSEp2Ym1sclFtbHMKWjJsS
GRYWmxibXhwWjJsQlUwMUxUa1ZUU1MxVE1pOU1ZWFJsYzNSRFVrd3VZM0pzTUE0R0ExVWREd0VCL3dRRQpBd0lHd0RDQmd3WUlLd1lCQlFVSEFRTUVkekI
xTUFnR0JnUUFqa1lCQVRCcEJndGdoaGdCUFFBQnAwNEJBUXhhClFuVWdjMlZ5ZEdsbWFXdGhMQ0ExTURjd0lITmhlV2xzYVNCRmJHVnJkSEp2Ym1scklFb
HRlbUVnUzJGdWRXNTEKYm1FZ1o4TzJjbVVnYm1sMFpXeHBhMnhwSUdWc1pXdDBjbTl1YVdzZ2MyVnlkR2xtYVd0aFpHbHlNRkFHQTFVZApDUVJKTUVjd0Z
BWUlLd1lCQlFVSENRSXhDQVFHUVc1cllYSmhNQjBHQ0NzR0FRVUZCd2tCTVJFWUR6RTVOemd4Ck1qTXhNakF3TURBd1dqQVFCZ2dyQmdFRkJRY0pCREVFQ
kFKVVVqQVlCZ05WSFJFRUVUQVBnUTFtYVhkbFFHWnAKY21VdVkyOXRNQjBHQTFVZERnUVdCQlNNQTBYZFNQbWhsc2tmVDVVOEdmL3dKZWgybVRBTkJna3F
oa2lHOXcwQgpBUXNGQUFPQ0FRRUFkOTErM1k2NGVDeWJyQ1VmM3I3R2tUQno1MkF3N3E1b09TeWlwVU1GbDNlamxXYVptUlVNCjdBemxhNU1qditGa21Nc
mxYWDNYZHNHYk9KVTFJMjdsNFpiUGVzVWwvdFJtSmVDMlJVeTMwQnlySXRTN3FXb00KWW4wbmh4TzN3MTJIQlc4dll1Nm1aUTdibkl6TytzODJQdk5QWFV
NYnBYZVNUMTU3djBnelJBcTJ3THRySFVjMwpGVE9ZaFl4czdJZmZnNWZvdDZWQUNTS2l0RkVhSWpiME9QT1lZNkJpMnFtRzJmRVpoMWJPbXFYWC96N3Vvd
HczCmZXWFdLd29OcmZhZms1dVIxWFVOSzJ3SnZSMHpCV1QxczJYK2YzOW4yN2lUSUhpdDI2MytWQXhJcTJFQ3pmb0kKRUR6VXhZeE8zV01vSXQ5b1g1ayt
1MWtMVVpYNFgveWdyUT09PC9YNTA5Q2VydGlmaWNhdGU+PC9YNTA5RGF0YT48L0tleUluZm8+</ns5:Base64Signature>
            </ns5:MSS_Signature>
            <ns5:SignatureProfile>
               <ns5:mssURI>http://uri.turkcellteknoloji.com.tr/MobileSignature/profile3</ns5:mssURI>
            </ns5:SignatureProfile>
            <ns5:Status>
               <ns5:StatusCode Value="500"/>
               <ns5:StatusMessage>MULTIPLE_SIGN_COMPLETED</ns5:StatusMessage>
            </ns5:Status>
         </MSS_SignatureResp>
      </ns1:MSS_SignatureResponse>
   </soap:Body>
</soap:Envelope>
```

6.2 MSS_StatusQuery

The MSS_StatusQuery operation takes an input of type MSS_StatusRequest and responds with MSS_StatusResponse.

6.2.1 MSS_StatusRequest

The fields that make up MSS_StatusRequest are shown on Table 6.

Table 6, MSS_StatusRequest fields

Name	Type	Description	Req
MajorVersion	Integer	This value is always ‘1’.	Y
MinorVersion	Integer	This value is always ‘1’.	Y
AP_Info			
AP_ID	anyURI	ID of the AP. This value is assigned by the MSSP Validation Service operator.	Y
AP_PWD	String	Password of the AP. This value is assigned by the MSSP Validation Service operator.	Y
AP_TransID	NCName	A transaction number, created by the AP.	Y
Instant	DateTime	The date and time when the message was created.	Y
MSSP_Info	MeshMemberType		Y
DNSName	String	Name of the MSSP if it is known by the AP. If it is not known, an empty value is also accepted.	Y
MSSP_TransID	NCName	A transaction number created by the MSSP Validation Service for this transaction.	Y

6.2.2 MSS_StatusResponse

The fields that make up MSS_StatusResponse are shown on Table 7. The MSS_Notification_Request that the MSSP sends to the AP as a response for signing also contains this message. See also MSS_NotificationRequest on chapter 6.5.

Table 7, MSS_StatusResponse fields

			Req
MajorVersion	Integer	This value is always ‘1’.	Y
MinorVersion	Integer	This value is always ‘1’.	Y
AP_Info			Y
AP_ID	anyURI	ID of the AP. This value is assigned by the TTech Msign Validation Service – MSSP operator.	Y
AP_TransID	NCName	A transaction number, created by the AP.	Y
Instant	DateTime	The date and time when the message that this message is a response to was created.	Y
MSSP_Info			Y
MSSP_ID	MeshMemberType	All APs in TTech Msign Validation Service – MSSP are identified by means of URI-based identifiers.	Y
Instant	DateTime	The date and time when the message was created.	Y
MobileUser	MobileUserType		Y
MSISDN	String	The end-user’s telephone number, for example +3581234567890.	Y
UserIdentifier	String	If the feature is configured in TTech Msign Validation Service – MSSP, this field can contain information read from the user certificate.	N
Status	StatusType,	Current status of the transaction. For more information, see chapter 0.	Y
MSS_Signature	SignatureType,	The signature.	N

6.3 MSS_ProfileQueryRequest

The fields that make up MSS_ProfileQueryRequest are shown on Table 8. MSS_ ProfileQueryRequest is a message for requesting end user certificate information before sending the signing request.

Table 8, MSS_ProfileQueryRequest fields

			Req
MajorVersion	Integer	This value is always '1'.	Y
MinorVersion	Integer	This value is always '1'.	Y
AP_Info			
AP_ID	anyURI	ID of the AP. This value is assigned by the TTech Msign Validation Service – MSSP operator.	Y
AP_PWD	String	Password of the AP. This value is assigned by the TTech Msign Validation Service – MSSP operator.	Y
AP_TransID	NCName	A transaction number, created by the AP.	Y
Instant	DateTime	The date and time when the message was created.	Y
MSSP_Info	MeshMemberType		Y
DNSName	String	The DNS name of the HomeMSSP if it is known by the AP. If it is not known, an empty value is also accepted.	Y
MSSP_TransID	NCName	A transaction number created by the MSSP for this transaction.	Y
Message	DataType	The message to be sent to the user. The data is assumed to be text, so the mime-type and encoding attributes are not required. Empty message are accepted but ignored empty messages.	Y

`CertHash` element of `SignatureProfile` element in `MSS_ProfileResp` is used to return the reference to the certificate used by the subscriber.

For convenience ASP can use `UserIdentifier` optional element of `MobileUser` element of `MSS_ProfileReq` to request for a digest method. The following digest method URIs are expected.

- `SHA1` – http://www.w3.org/2000/09/xmldsig#sha1
- `SHA256` – http://www.w3.org/2001/04/xmlenc#sha256
- SHA384 – http://www.w3.org/2001/04/xmldsig-more#sha384
- SHA 512 – http://www.w3.org/2001/04/xmlenc#sha512

An example Profile Query is shown below

```
<?xml version="1.0" encoding="utf-8"?>
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  <soapenv:Body>
    <MSS_ProfileQuery xmlns="">
      <MSS_ProfileReq MajorVersion="1" MinorVersion="1">
        <ns1:AP_Info AP_ID="http://MImzaTest"
        AP_PWD="ABC123" AP_TransID="_20121017150222470"
        Instant="2012-10-17T12:02:22.470Z"
        xmlns:ns1="http://uri.etsi.org/TS102204/v1.1.2#" />
        <ns2:MSSP_Info Instant="2012-10-17T12:02:29.471Z"
        xmlns:ns2="http://uri.etsi.org/TS102204/v1.1.2#">
          <ns2:MSSP_ID />
        </ns2:MSSP_Info>
        <ns3:MobileUser xmlns:ns3="http://uri.etsi.org/TS102204/v1.1.2#">
          <ns3:UserIdentifier>http://www.w3.org/2001/04/xmlenc#sha256</ns3:UserIdentifier>
          <ns3:MSISDN>05335935505</ns3:MSISDN>
        </ns3:MobileUser>
      </MSS_ProfileReq>
    </MSS_ProfileQuery>
  </soapenv:Body>
</soapenv:Envelope>
```

If a valid digest method is given in `UserIdentifier` element of type String,

- `SignatureProfile` → `CertHash` → `DigestMethod` value will be the given method
- `SignatureProfile` → `CertHash` → `DigestValue` value will be calculated with the given method

If an unrecognized digest method is received, then SOAP Fault Code `101 – WRONG_PARAM` is returned.

If no digest method is given in `UserIdentifier` element, then the following default values will be used. In every case AP should be able to parse `DigestMethod` and interpret `DigestValue` accordingly.

Applet Version	SignatureProfile → mssURI	DigestMethod
1	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile1	http://www.w3.org/2000/09/xmldsig#sha1
2	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile2	http://www.w3.org/2001/04/xmlenc#sha256
3	http://uri.turkcellteknoloji.com.tr/MobileSignature/profile3	http://www.w3.org/2001/04/xmlenc#sha256

6.4 MSS_ProfileQueryResponse

The fields that make up MSS_ProfileQueryResponse are shown on Table 9. MSS_ ProfileQueryResponse is a reply from MSSP Validation Service. This reply is received as a response to the MSS_ ProfileQueryRequest.

Table 9, MSS_ReceiptResponse fields

			Req
MajorVersion	Integer	This value is always '1'.	Y
MinorVersion	Integer	This value is always '1'.	Y
AP_Info			Y
AP_ID	anyURI	ID of the AP. This value is assigned by the TTech Msign Validation Service – MSSP operator.	Y
AP_TransID	NCName	A transaction number, created by the AP.	Y
Instant	DateTime	The date and time when the message that this message is a response to was created.	Y
MSSP_Info			Y
MSSP_ID	MeshMemberType	All APs in TTech Msign Validation Service – MSSP are identified by means of URI-based identifiers.	Y
Instant	DateTime	The date and time when the message was created.	Y
MobileUser	MobileUserType		Y
MSISDN	String	The end-user's telephone number, for example +9053234567890.	Y
UserIdentifier	String	If the feature is configured in TTech Msign Validation Service – MSSP, this field can contain information read from the user certificate.	N
Status	StatusType,	Current status of the transaction. For more information, see chapter 0.	Y

A sample ProfileQueryResponse is shown below

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope">
   <soap:Body>
      <ns1:MSS_ProfileQueryResponse xmlns:ns1="http://turkcelltech.com/mobilesignature/validation/soap">
         <MSS_ProfileResp MajorVersion="1" MinorVersion="1" xmlns:ns2="http://www.w3.org/2000/09/xmldsig#"
xmlns:ns3="http://www.w3.org/2003/05/soap-envelope" xmlns:ns4="http://www.w3.org/2001/04/xmlenc#"
xmlns:ns5="http://uri.etsi.org/TS102204/v1.1.2#">
            <ns5:AP_Info AP_ID="http://xxx.com.tr" AP_TransID="_12647*****" AP_PWD="xxx123"
Instant="2020-02-13T12:49:36.832+02:00" AP_URL=""/>
            <ns5:MSSP_Info Instant="2020-02-19T13:23:23.211+02:00">
               <ns5:MSSP_ID/>
            </ns5:MSSP_Info>
            <ns5:SignatureProfile>
               <ns5:mssURI>http://uri.turkcellteknoloji.com.tr/MobileSignature/profile3</ns5:mssURI>
               <ns5:CertIssuerDN>C=TR,O=Elektronik Bilgi Guvenligi A.S.,CN=TEST Turkcell Mobil NES Hizmet Saglayicisi
S2</ns5:CertIssuerDN>
               <ns5:CertSerialNumber>617500238946994417636986445319925200 21</ns5:CertSerialNumber>
               <ns5:CertHash>
                  <ns5:DigestMethod Algorithm="http://www.w3.org/2001/04/xmlenc#sha256"/>
                  <ns5:DigestValue>/1HaiEPi8Do7ui4dJrjsO0DW0gepii6c7EWcnA91b4U=</ns5:DigestValue>
               </ns5:CertHash>
               <ns5:msspUrl>http://localhost</ns5:msspUrl>
               
<ns5:certIssuerDN-DER>MG8xCzAJBgNVBAYTAlRSMSgwJgYDVQQKDB9FbGVrdHJvbmlrIEJpbGdpIEd1dmVubGlnaSBBLlMuMTYwNAYDVQQDDC1URVNU
IFR1cmtjZWxsIE1vYmlsIE5FUyBIaXptZXQgU2FnbGF5aWNpc2kgUzI=</ns5:certIssuerDN-DER>
            </ns5:SignatureProfile>
            <ns5:Status>
               <ns5:StatusCode Value="100"/>
               <ns5:StatusMessage>REQUEST_OK</ns5:StatusMessage>
            </ns5:Status>
         </MSS_ProfileResp>
      </ns1:MSS_ProfileQueryResponse>
   </soap:Body>
</soap:Envelope>
```

6.5 MSS_NotificationRequest

The MSS_NotificationRequest is a message that MSSP Validation Service sends to the client in Asynchronous Server-Server mode. MSS_NotificationRequest is identical to MSS_StatusResponse message.

6.6 MSS_NotificationResponse

The MSS_NotificationResponse is a message that MSSP Validation Service expects to receive from the customers system as a response. MSSP Validation Service accepts this as a successful response, but does not send anything to the end-user.

7 Signature & Profile Service URLs

Prod Env

signatureServiceUrl	https://msign.turkcell.com.tr/MSSP2/services/MSS_Signature?wsdl
profileServiceUrl	https://msign.turkcell.com.tr/MSSP2/services/MSS_ProfileQueryPort?wsdl

8 Abbreviations

Terms, acronyms and Abbreviations defined on [MSSP] are valid on this document.

	Explanation
AP	Application Provider. AP is the party that needs the end user's signature.
API	Application programming interface. An API is an interface that a computer system or application provides to allow service requests to be made of it by other computer programs, and/or to allow data to be exchanged between them.
CA	A certificate authority or certification authority (CA) is an entity which issues digital certificates for use by other parties. It is an example of a trusted third party.
Certificate	A public key certificate is a certificate that uses a digital signature to bind together a public key with an identity — information such as the name of a person or an organization, their address, and so forth. The certificate can be used to verify that a public key belongs to an individual.
DN	Distinguished names (DNs) are used to uniquely identify entries in an LDAP or X.500 directory. DNs are strings typically used whenever you must add, modify or delete an entry in a directory using the LDAP programming interface.
ETSI	The European Telecommunications Standards Institute (ETSI) is an independent, non-profit, standardization organization of the telecommunications industry (equipment makers and network operators) in Europe.
MSISDN	The Mobile Station Integrated Services Digital Network (MSISDN) refers to the 15-digit number that is used to refer to a particular mobile station.
OTA	Over The Air (OTA) is a standard for the transmission and reception of application-related information in a wireless communications system.
PKCS	The Public-Key Cryptography Standards (PKCS) are a set of inter-vendor standard protocols for making possible secure information exchange on the Internet using a public key infrastructure (PKI). See also PKI.
PKCS#1	RSA Cryptography Standard defining the format of RSA encryption. See also PKCS.
PKCS#7	This standard defines the cryptographic message syntax, which is used to sign and/or encrypt messages under a PKI. See also PKCS.
PKI	A public key infrastructure (PKI) enables users of a basically insecure public network such as the Internet to securely and privately exchange data by using a public and a private cryptographic key pair obtained and shared through a trusted authority.
RPC	Remote procedure call (RPC) is a protocol allowing a computer program running on one computer to execute a subroutine on another computer without the programmer explicitly coding the details for this interaction. In object-oriented programming terminology, RPC may be referred to as remote invocation or remote method invocation.
SDK	A software development kit (SDK) is a set of development tools that allows a software engineer to create applications for a certain software package.
SIM	A SIM card, also known as a subscriber identity module, is a smart card that stores data for GSM cellular telephone subscribers.
SOAP	SOAP is a protocol for exchanging XML-based messages over a computer network, typically using HTTP. SOAP forms the foundation layer of the Web services stack by providing a basic

9 References

Table 10, References

[RFC2119]	"Key words for use in RFCs to Indicate Requirement Levels". S. Bradner. March 1997. URL:http://www.ietf.org/rfc/rfc2119.txt
[MSSP]	See [204], [207] and [206]
[203]	ETSI TR 102 203 V1.1.1 (2003-05) Technical Report Mobile Commerce (M-COMM); Mobile Signatures; Business and Functional Requirements
[204]	ETSI TS 102 204 V1.1.4 (2003-08) Technical Specification Mobile Commerce (M-COMM); Mobile Signature Service; Web Service Interface
[206]	ETSI TR 102 206 V1.1.3 (2003-08) Technical Report Mobile Commerce (M-COMM); Mobile Signature Service; Security Framework
[207]	ETSI TS 102 207 V1.1.3 (2003-08) Technical Specification Mobile Commerce (M-COMM); Mobile Signature Service; Specifications for Roaming in Mobile Signature Services
[MSS]	See [MSSP].
[PKCS#7]	PKCS #7: Cryptographic Message Syntax Standard, An RSA Laboratories Technical Note, Version 1.5, Revised; November 1, 1993.
[PKCS#1]	PKCS #1 v2.1: RSA Cryptography Standard, RSA Laboratories; June 14, 2002
[RFC3275]	(Extensible Markup Language) XML-Signature Syntax and Processing; Internet Engineering Task Force (IETF), Network Working Group, Request for Comments: 3275, Obsoletes: 3075, Category: Standards Track; D. Eastlake 3rd Motorola, J. Reagle W3C, D. Solo Citigroup; March 2002